

Prompt Experiments

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can explain the mechanism and terminology of controlled prompt experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

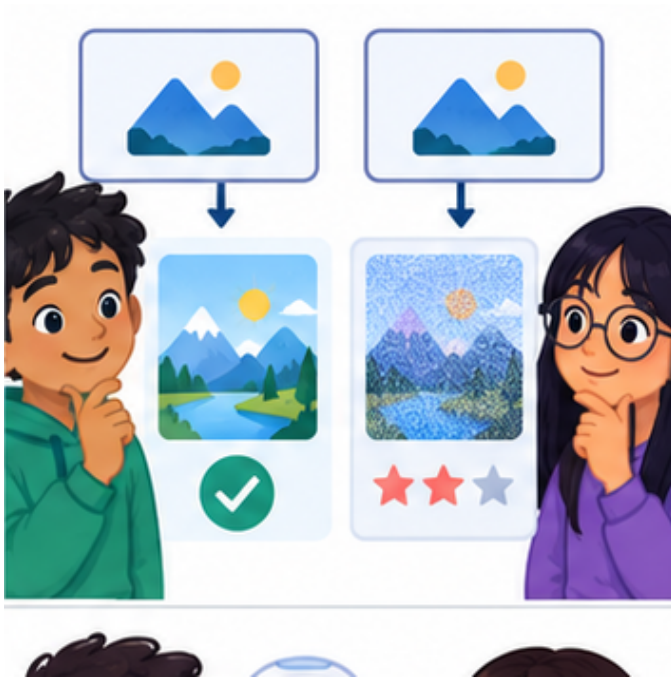
Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 28: Prompt Experiments

Prompt quality is evaluated by controlled comparisons and success criteria.



THINKER MISSION

Change one prompt variable at a time and score results.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Structured outputs and validation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

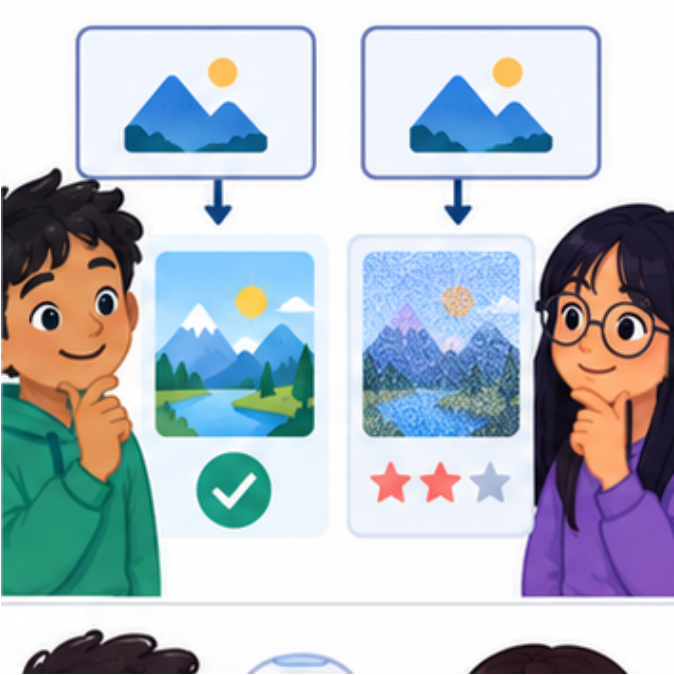
By the end, I can compare prompts on paired cases and inspect a subgroup before choosing.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use prompt experiments to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings.

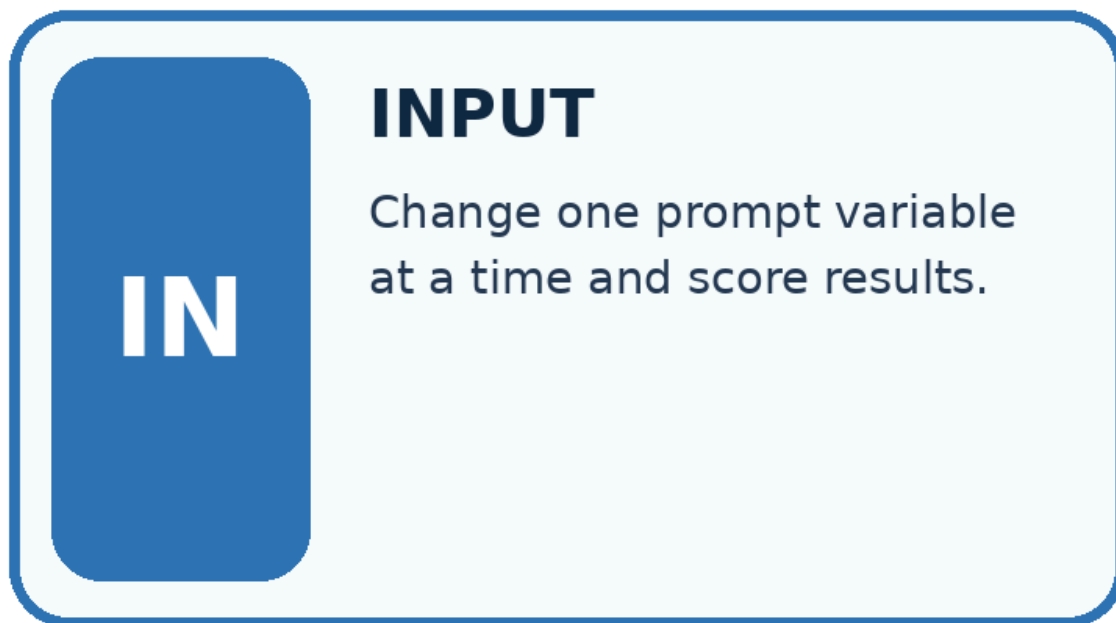
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Prompt quality is evaluated by controlled comparisons and success criteria.



OUT

OUTPUT

Change one prompt variable at a time and score results.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Controlled prompt experiments — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Check or calculation: Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.

Conclusion and limit: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Make missing-source correctness a required selection gate.

Prediction: Neither prompt clears a 2/2 gate on these records.

Reason: P fails 7 and Q fails both 4 and 7; both need revision.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

PROMPT

instruction or input to a
generative system

2

CONSTRAINT

required boundary

3

CRITERION

rule for judging success

4

CONTROLLED COMPARISON

only one deliberate
difference

Word · My own example · Not this / boundary

Word: prompt

My own example: Input instructions and context supplied to a generative system.

Not this / boundary: _____

Word: constraint

My own example: A condition that a valid solution must satisfy.

Not this / boundary: _____

Word: criterion

My own example: One stated standard used to judge a result.

Not this / boundary: _____

Word: controlled comparison

My own example: A comparison designed to keep relevant conditions alike except for the factor being studied.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Paired case

Meaning: The same case evaluated under each compared procedure

Word: Rubric

Meaning: Prespecified criteria for judging an output

Word: Regression

Meaning: A previously passing condition that fails after a change

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 28 • WORKED EVIDENCE

- 1 Prompt A meets 3 of 6 rubric checks.
- 2 Add one constraint: 'Use 3 sentences and name no ending.'
- 3 Prompt B meets 5 of 6 checks.
- 4 Improvement = +2 checks; repeat on more books before claiming it generalizes.

1. Prompt A meets 3 of 6 rubric checks.
2. Add one constraint: 'Use 3 sentences and name no ending.'
3. Prompt B meets 5 of 6 checks.

4. Improvement = +2 checks; repeat on more books before claiming it generalizes.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Experiments

A fictional book-help project tests two prompt versions on seven shared development cases.

The facts we will use

P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same inputs, system configuration and rubric; only the prompt changes. These are invented records.

Paired overall and missing-source passes

	Overall passes /7	Missing-source passes /2
P	4	1
Q	5	0

Read a cell using both its row and column labels.

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

$P=4/7$; $Q=5/7$.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Paired setup

What happens and why: Use the same seven cases and pass criteria for both prompts.

Step: 2. Overall rates

What happens and why: $P=4/7$; $Q=5/7$.

Step: 3. Missing-source slice

What happens and why: P passes 1/2; Q passes 0/2 in cases 4 and 7.

Step: 4. Decide explicitly

What happens and why: Q improves overall count but worsens missing-source behavior; use the declared priorities.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection. Keep the failing examples for development and obtain fresh cases after choosing a final version.

What this example does not establish

Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Read the labelled picture: Name the case P passes and Q fails, and explain why it matters.

Change one thing: Make missing-source correctness a required selection gate. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: One impressive output proves a prompt is reliable. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Prompt selection on development cases cannot be reported as independent final performance.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled prompt experiments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Name the case P passes and Q fails, and explain why it matters.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Define six rubric checks, keep task and examples fixed, change one prompt element, score both versions across three cases, and explain uncertainty.

Define six rubric checks, keep task and examples fixed, change one prompt element, score both versions across three cases, and explain uncertainty.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER
Test generated book explanations with a fixed rubric and prohibit invented details.

THIS WEEK'S CHECKPOINT Test generated book explanations with a fixed rubric and prohibit invented details.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

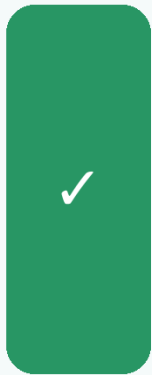
Run controlled recommender explanation experiments with paired cases and report important failure conditions separately.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Change one prompt variable at a time and score results.

CHALLENGE Change one prompt variable at a time and score results.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Prompt B adds under 100 pages and explain the match to Prompt A. Both use a catalogue.

The question

Design a fair comparison.

Worked explanation

Use identical requests, catalogue and scoring criteria. Check constraint compliance and supported explanation; repeat stochastic outputs when relevant.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare prompts under the same conditions

Evidence or stage · Value or result

Evidence or stage: Prompt A

Value or result: Original catalogue request

Evidence or stage: Prompt B

Value or result: Add under 100 pages and explain the match

Evidence or stage: Hold fixed

Value or result: Catalogue, requests and scoring

Evidence or stage: Check

Value or result: Constraint compliance and supported explanation

Prompt B adds under 100 pages and explain the match to Prompt A. Both use a catalogue.

Design a fair comparison.

Use identical requests, catalogue and scoring criteria. Check constraint compliance and supported explanation; repeat stochastic outputs when relevant.

14. Try a new case independently

Independent case: Apply a new case independently

On eight paired cases A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a controlled comparison.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Test generated book explanations with a fixed rubric and prohibit invented details.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when outputs vary across runs.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional book-help project tests two prompt versions on seven shared development cases.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of controlled prompt experiments

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of controlled prompt experiments

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-help project tests two prompt versions on seven shared development cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Q improves overall count but worsens missing-source behavior; use the declared priorities.

1. Decide explicitly
2. Paired setup
3. Missing-source slice
4. Overall rates

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Decide explicitly Q improves overall count but worsens missing-source behavior; use the declared priorities. A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection. Keep the failing examples for development and obtain fresh cases after choosing a final version.

Practice 2: explain the deciding evidence

Changed-case practice: Make missing-source correctness a required selection gate. Which conclusion is

supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Neither prompt clears a 2/2 gate on these records.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Neither prompt clears a 2/2 gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Repair a missing building block

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank training neighbours. Short practice: Why do $k=1$ and $k=3$ disagree for the original query?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.